

Agronomic Insights

The comprehensive analysis reveals several critical patterns that provide deeper understanding of rice yield determinants:

Model Robustness and Meteorological Dominance The consistently high R^2 scores (≈ 0.75) across all Random Forest configurations demonstrate robust predictive capability, while the extremely high training R^2 (0.962) indicates the model captures complex interactions effectively. Meteorological factors show 2.5 times greater average importance than soil chemistry factors, confirming that climate variables are the primary yield drivers in this system.

Growth Stage-Specific Radiation Requirements The dramatic difference between Radiation4 (grain filling period, SHAP value ~ 0.14) and other radiation periods reveals rice's critical dependence on solar energy during grain filling. This aligns with the physiological reality that 70-80% of final grain weight comes from current photosynthesis during this period, not stored carbohydrates. The lower importance of early-stage radiation suggests that vegetative growth can compensate for suboptimal early conditions, but grain filling cannot.

Temperature Thresholds and Phenological Sensitivity Maximum temperature during early growth (Max temperature1) shows the highest temperature importance, indicating that adequate heat units during tillering and stem elongation are crucial for establishing yield potential. The correlation matrix reveals strong positive relationships between temperature variables across growth stages ($r \approx 0.96-1.00$), suggesting that seasonal temperature patterns, rather than isolated temperature events, drive yield responses.

Soil Chemistry Optimization Windows The organic matter content and carbon-nitrogen ratio show both positive and negative SHAP values, indicating optimal ranges exist. The moderate correlation between these variables ($r = 0.21$) suggests they can be managed somewhat independently. Available phosphorus consistently shows positive impacts without negative values, indicating this system may be phosphorus-limited across the range studied.

Water-Nutrient Interactions The precipitation patterns across growth stages show varying importance, with Precipitation4 (likely monsoon period) being most critical. The correlation between electrical conductivity and various soil parameters suggests

complex soil-water-nutrient interactions that affect nutrient availability during different moisture conditions.

Specific Recommendations for Farmers

Strategic Planting Date Optimization Calculate planting dates to position grain filling (35 days post-heading) during peak solar radiation months. Use local weather data to identify the 30-day period with highest average daily radiation and count backward approximately 100-120 days depending on variety maturity. This single adjustment can capture the most influential yield factor.

Precision Temperature Management Install simple max-min thermometers in fields to track temperature accumulation during early growth. Target transplanting when consistent daily maximum temperatures exceed 28°C for optimal tillering. In areas with unpredictable spring weather, use protective measures like plastic tunnels during the first 30 days post-transplanting.

Soil Organic Matter Enhancement Maintain soil organic matter between 2.5-3.5% through strategic residue management. Incorporate rice straw immediately after harvest but ensure C:N ratios of 25:1 or lower by adding nitrogen fertilizer (20-30 kg N/ha) during incorporation. This prevents nitrogen immobilization while building long-term soil health.

Phosphorus Status Monitoring and Management Conduct annual soil phosphorus testing and maintain available P levels above 15 mg/kg through targeted applications. Apply phosphorus fertilizers as basal dressing before transplanting when soil temperatures are rising, enhancing root development during the critical establishment period.

Water Management for Nutrient Optimization Implement alternate wetting and drying (AWD) during vegetative growth to concentrate nutrients and improve root development, but maintain continuous flooding during grain filling to maximize photosynthetic efficiency. Monitor electrical conductivity weekly and flush fields if EC exceeds 2.0 dS/m during sensitive growth phases.

Variety Selection Based on Local Climate Patterns Choose varieties with 110-130 day maturity that can complete grain filling before seasonal radiation decline. In areas with variable weather, select varieties with proven tolerance to temperature

fluctuations during reproductive phases, prioritizing heat tolerance over cold tolerance based on the temperature importance patterns observed.

Integrated Nutrient Timing Strategy Split nitrogen applications to coincide with temperature-driven growth phases: 40% at transplanting, 30% at maximum tillering (when temperatures peak), and 30% at panicle initiation. This synchronizes nutrient availability with the plant's temperature-enhanced metabolic capacity.

Risk Management for Weather Variability Establish field-level weather monitoring to track radiation accumulation during grain filling. If radiation falls below 150 MJ/m^2 during the 35-day post-heading period, consider supplemental lighting in high-value plots or adjust harvest timing to minimize quality losses from incomplete grain filling.

These recommendations work synergistically - optimal planting dates enhance the effectiveness of nutrient management, while proper soil preparation maximizes the benefits of favorable weather conditions during critical growth periods.